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ECO-WORTHY
Make Rural Life Better

A black and white photograph of two people on a boat. One person is standing and holding a long pole, while the other is sitting. The boat is equipped with solar panels on its deck. The image is tilted at an angle.

10W 25W SOLAR PANEL KIT

SUPPORT

If you are experiencing technical problems and cannot find a solution in this manual, please contact ECO-WORTHY for further assistance.

Tel: +44 20 7570 0328(EU)
1-866-939-8222(US)

E-mail: customer.service@eco-worthy.com
Web: www.eco-worthy.com

Contents

Product Introduction	1
Package Included	1
Installation of solar panel kit.....	2
Application	8
Caution	9
Data Sheet	9
Trouble Shooting.....	10
Warranty	11

Dear Customer, Thank you for selecting this ECO-WORTHY solar panel kit. This manual explains the installation of your solar kit and answers some frequently asked questions. Please read this manual carefully before setting up your system. Remember, all work should be carried out by an appropriately qualified person. Precautions and safety measures should be taken in all cases.

If you have any concerns about the suitability of the kit for your application, or doubts about any of the instructions in this manual, please contact our Support at +44 20 7570 0328(EU) ,1-866-939-8222(US) or email customer.service@eco-worthy.com

Product Introduction

The 10W solar panel kit,using a 10W solar panel with 1M extension cable, a 10A PWM charge controller,a pair of 2m battery-controller cable with crocodile clips.It's a great choice for 12V battery charging,work for Vehicle ,Boat,Gate opener,Chicken coop battery charge.

The 25W solar panel kit,using a 25W solar panel, a 10A PWM charge controller, 2 battery cables with ring terminal and crocodile clip, for the convenience of wiring. And we use SAE connector to make solar panel kit connection more easier, plug and play!

Package Included

10W solar panel kit

- 10W Monocrystalline solar panel with 1m cable
- 10A PWM solar charge controller
- 2m controller-battery cable with crocodile clips

25W solar panel kit

- 25W Polycrystalline solar panel with 3m cable
- 10A PWM solar charge controller with SAE connector
- 40cm controller-battery cable with ring terminal
- U-shaped bracket

Installation of solar panel kit

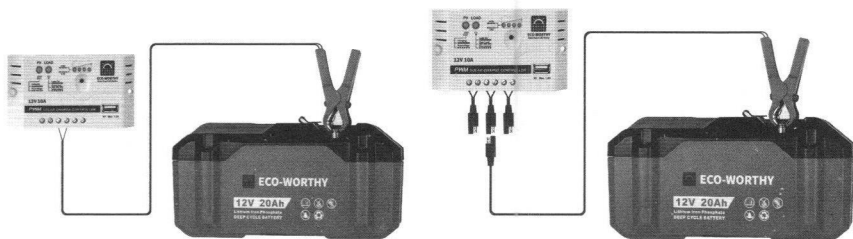
Before installing the solar kit, please ensure that your 12V battery is not overdischarge. The battery voltage should higher than 8V - this is the minimum voltage required to start the solar charge controller. If your battery is overdischarge, please charge it beforehand by other means for a short period of time.

Connect your 12V battery to the solar charge controller using the cable in the following step: (Left: Installation of 10W kit Right: Installation of 25W kit)

3-1 On the solar controller end, use a connector to connect the end of the cable



3-2 On the battery end, attach the terminal using the crocodile clips.



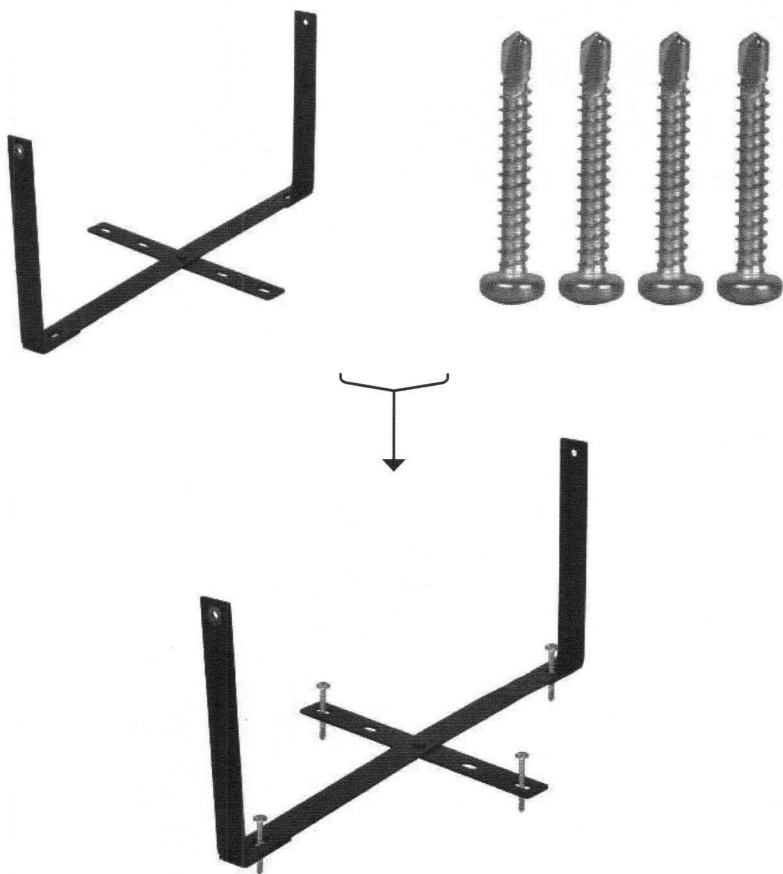
The RED battery terminal should be connected to the “+” battery terminal of the controller, and the BLACK battery terminal should be connected to the “-” battery terminal of the controller.

3-3 Install the U-bracket for 25W solar panel kit.

In the 25W solar panel kit package, there's a U-shaped bracket that for 25W solar panel only. Install it according to the following instructions.

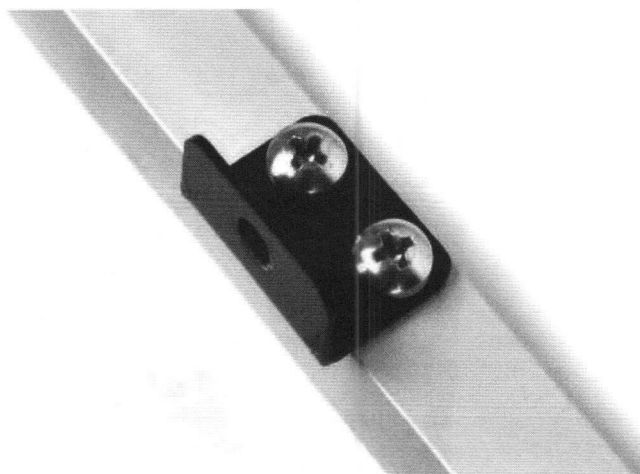
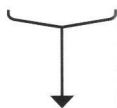
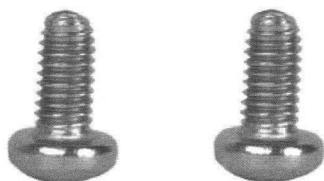
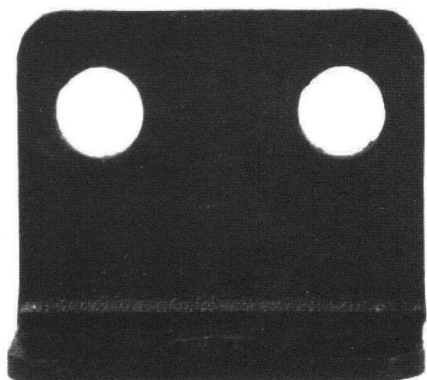
Step 1: Fix the U-shaped bracket on flat surface

Extend the middle cross rod of this U-type bracket, then put the U-type bracket on flat metal or wood surface (expansion plug needed for concrete) as the below image. Use 4 M4.8*32 Phillips pan head drill screws to fix the four sides tightly into the flat surface.



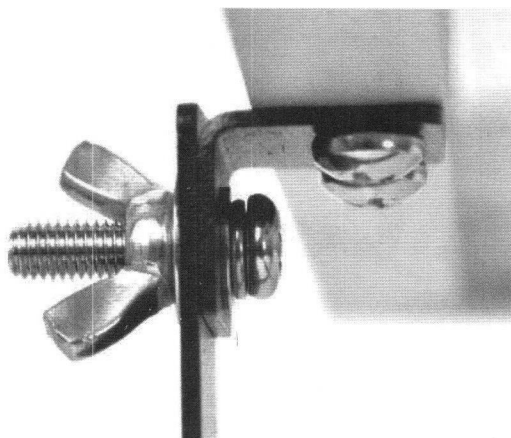
Step 2: Install connection plate on solar panel frame

There are two L-connecting pieces for frame connection, use 2 M5*10 Phillips pan head self-tapping screws to fix them on solar panel frame as below.
(Note: the screws needs to be drilled hard for the first installation)



Step 3: Connect the solar panel and the U-shaped bracket

Use the M5*20*9 screws and wing nuts to fix the two black plates on the U-type bracket.

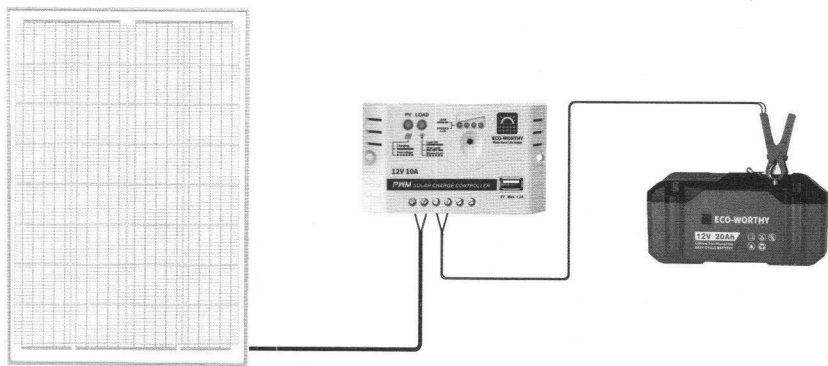


After adjusting the solar panel manually to the proper angle, tighten the wing nut.

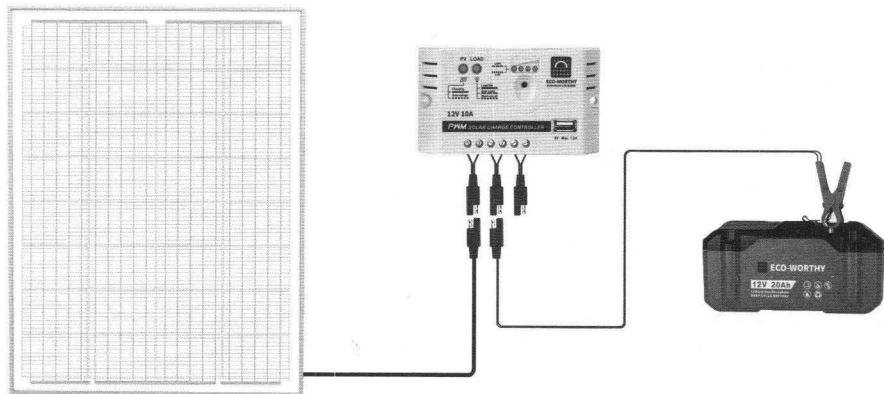
The whole 25W solar panel kit should look like this when fully assembled.



3-4 Use the connector's extension cable to connect the solar panel to the solar charge controller.



(10W Solar panel kit)



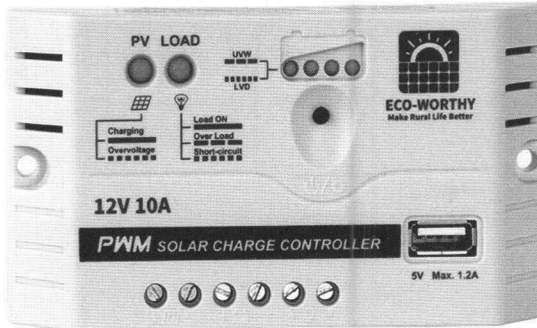
(25W Solar panel kit)

Battery type setting

Step 1: Enter setting mode by pressing button for 5s until the battery status LEDs are flashing.

Step 2: Select the desired mode by pressing button.

Step 3: The mode will be saved automatically without any operation for 5S and LED will stop flashing.

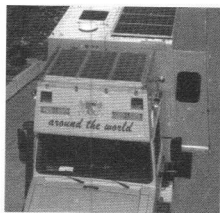


LED 1	LED 2	LED 3	Battery Type
●	○	○	Sealed (Default)
●	●	○	Gel
●	●	●	Flooded

- Important notes: If you want to charge your lithium battery with this charge controller, we would recommend you to select "Gel Battery Mode" for charging.

Application

Perfect for off grid 12 volt batteries charging system and a variety of DC applications, designed for charging 12 volt batteries in car, trucks, RVs, cabins, boats, marines, shed, cabin, gate opener, battery , chicken coop, home security system battery



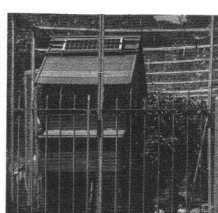
Car



Boats



Gate opener



Chicken coop

Caution

- For maximum output, ensure that the solar panel is situated outdoors and is exposed to as much sunlight as possible, without any shading. Even a small shaded area can significantly reduce the output (e.g. leaves, trees etc).
- Positioning the solar panel indoors behind a window is not recommended as it will reduce the output considerably.
- If required, you can connect any 12V load or appliances to the load terminals of the solar controller ("+" to "+", "-" to "-"), for example 12V light bulbs. Ensure that the total current drawn by the load does not exceed 10A.
- Your 12V appliances can be connected directly to your battery but discharge protection function of controller will be unavailable.

Data Sheet

Solar Panel		
Rated Power	10W	25W
Solar Cell	Monocrystalline	Polycrystalline
Maximum/Peak Voltage(Vmp)	18V	18V
Open Circuit Voltage(Voc)	22.41 V	22.41 V
Short circuit current	0.61A	1.54A
Maximum current	0.56A	1.4A
Temperature range	-40°C ~ +85°C	-40°C ~ +85°C
Frame	Aluminum alloy frame	Aluminum alloy frame
Length of extension cable	100cm(3.28ft)	300cm(9.84ft)
Size of extension cable	0.75mm ²	0.75mm ²
Size	337x200x18mm (13.3 x 8.1 x 0.7 in)	420x320x17mm (16.5 x 12.6 x 0.7 in)

10A 12V Charge Controller	
Nominal output(automatic identification)	12V
Rated charge/discharge current	10A
Battery input voltage range	8V~16V
Max. PV open circuit voltage	30V
Self-consumption	≤5mA
USB Output	5V 1.2A
Working environment temperature	-35℃ ~ +50℃
Overall dimension	120.3x67x21.8mm
Mounting dimension	111.5mm
Mounting hole size	Φ4.5
Terminals	12AWG/4mm ²
Net weight	0.10kg

Trouble Shooting

6-1 Check Wiring Terminations

- Ensure that all wire terminations are tight.
- Make sure no corrosion is present.
- Make sure wires are not chaffed.

6-2 Load Verification – Is there a voltage at the load? If there is no voltage, measure voltage on each piece of equipment leading back to the solar array until you find where the voltage and/or current drops.

6-3 Verify all interconnections between the batteries are tight & corrosion free.

If the battery voltage is between 11.5V and 12.6V , check the PWM controller battery status LEDs. Consult the charge controller manual for other LED colors or flashing sequences.

Check for any signs of swelling, shrinkage, or case cracks, these may indicate battery damage.

6-4 Solar panel – The two most important numbers when troubleshooting the array are the Short Circuit current (Isc) and Open Circuit voltage (Voc). Refer to module data sheet for both values.

Voc – The Voc should be fairly consistent between each module in your array and should be around 20-22VDC for a 12V solar panel

Isc – The short circuit current (Isc) should ideally be tested on a sunny day around solar noon.

6-5 Using a Multimeter – Once all the above issues and concerns have been addressed, it's time to begin measuring electrical parameters (voltage and current) on each side of our electrical devices to look for any faulty components. With all components on, we will attempt to trace the voltage from the load to the solar array, through each electrical device.

Check voltage on load side and battery side of Load Controller

Check voltage of battery

Check voltage on battery side of charge controller

Check voltage on solar panel side of charge controller

Warranty

This product is covered by a 1 year warranty provided by **ECO-WORTHY Ltd.** We will refund or a part of refund or replace any products with defects at our discretion.



Disclaimer

Working with electricity and batteries can be dangerous. The information provided in this manual is for general guidance only. All work should follow the safety standards and should be carried out by an appropriately qualified person.

Eco-worthy Ltd. is not responsible for any damage or injury caused by inappropriate installation or use of the product



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Make Rural Life Better



10W 25W SOLARPANEL-KIT

UNTERSTÜTZUNG

Wenn Sie technische Probleme haben und in diesem Handbuch keine Lösung finden, wenden Sie sich bitte an ECO-WORTHY, um weitere Unterstützung zu erhalten.

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